

AIM

PT.010 PERFORMANCE TESTING STRATEGY

<Company Long Name>

<Subject>

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Introduction

Purpose

The purpose of the **Performance Testing Strategy** document is to provide a reference at any time for resources working on the *Performance Testing* process of the <Project Name>. This document is also the source for performance testing approaches, direction, risks, benefits, and assumptions.

Performance Testing Strategy covers the delivery of project and support services to assist the company in the completion of the Performance Testing subproject. <Performance Testing Team Leader> will direct the Performance Testing subproject, with the assistance of internal project team members as necessary. The purpose of this document is to provide the performance testing subproject team a single reference for:

- Performance Testing scope, objectives, milestones, and critical success factors
- Performance Testing approaches: management, environment, and other business initiatives
- Project policies, risks, and assumptions
- Acceptance criteria and change management
- Overall strategy for conducting the performance testing activities
- Potential impacts of performance testing on the overall project

Distribution of this document to all performance testing team members will occur at the subproject kickoff meeting. Communication of any subsequent changes to the project scope will occur universally before implementation of these changes commences.

Background

The overall <Project Name> implementation project includes the performance-testing subproject.

The goal of the main <Project Name> implementation project is to...

The goal of the performance testing subproject is to...

Scope and Application

This **Performance Testing Strategy** document covers the Performance Testing subproject, which is a component of the <Project Name> project.

Related Documents

Project Management Plan

This Performance Testing strategy expands the <Project Name> Project Management Plan with greater detail for the Performance Testing components of the overall implementation project.

Sources

Preparation of this **Performance Testing Strategy** document resulted from discussions and input from the following sources:

-
-
-

Scope

Scope of Subproject

General Scope Areas

The general scope of the performance test subproject includes a number of areas:

Scope Area	Included?	Description
Applications	✓	New Oracle Applications installation will be performance tested
Technical Architecture	✓	The technical architecture, including applications server configuration and network, will be performance tested
Interfaces	☐	Interfaces will not be performance tested
Database	✓	The performance of the database will be tested
Business Processes	☐	Full end-to-end business processes will not be performance tested
Business Functions	✓	Discrete business functions that have been identified as risks will be tested
Manual or Automated Testing	✓	The Performance Test will be conducted manually

Scope In Detail

Applications

The following table lists the applications considered to be *within* scope for the performance testing project:

Application Category	Application	Abbreviation
Application Technology	Application Object Library	FND
Application Technology	Alert	ALR
Finance	General Ledger	GL

Application Category	Application	Abbreviation

Technical Architecture

The technical architecture aspects considered to be *within* scope for the performance testing process are:

Architecture Component	Measurements	Description

Business Functions

The following table lists the business functions considered to be *within* scope for the performance testing project:

Business Unit (Organization/Site)	Business Function	Description

Milestones

There are distinct certification points throughout a project. These points usually follow project milestones and phase completion. Project management will approve each project milestone or deliverable by using a standard Acceptance Certificate as described in the Project Management Plan.

The architecture subproject team has defined a number of such milestones. The steering committee and project management may use these milestone target dates to evaluate progress towards major deliverables:

- **Definition Phase Complete** – Performance testing scope and strategy defined
- **Operations Analysis Phase Complete** – Performance testing models defined
- **Solution Design Phase Complete** – Performance test database and transaction programs designed

- **Build Phase Complete** – Performance test executed and documented

The formal review and approval of these project milestones is the responsibility of:

-
-
-

Approvers will utilize the standard acceptance criteria for each task deliverable produced, in order to assess deliverable completeness.

The following are the detailed milestones defined by the Performance Testing subproject team:

Phase	Milestone	Date	Primary Responsibility
Definition	Performance Testing Strategy Complete		
Operations Analysis	System Models Complete		
Solution Design	Transaction Program Designs Complete		
	Test Database Designs Complete		
Build	Database Population Complete		
	Transaction Programs Complete		
	Test Environment Complete		
	Test Execution Complete		
	Test Report Complete		

Constraints and Assumptions

The following is a list of identified project constraints:

- Hardware Availability
- Software Release Versions and Availability
- Automated Testing Tool Availability
- Budget Constraints
- Resource Availability

Hardware Availability

<Describe hardware availability constraints>

Software Release Versions and Availability

<Describe software release versions and availability constraints>

Automated Testing Tool Availability

<Describe project constraints arising from testing tool absence or limitations>

Budget Constraints

<Describe budget constraints>

Resource Availability

<Describe resource constraints, particularly expert resources>

Risks

The following are the risks identified that may affect the project during its progression:

-
-
-

Tracking of these and any other risks identified later, follows through to the Risk and Issue Management process defined in the Project Management Plan.

Scope Control

Management control of changes to the scope identified in this document follows through to the Change Management procedure defined in the main Project Management Plan. We will use Change Request Forms to identify and manage changes, with approval for any changes that affect cost or time scale for this subproject.

Relationship to Other Systems/Projects

It is the responsibility of <Company Short Name> to inform the performance testing team of other business initiatives that may impact the project. The following is a list of business initiatives that may impact performance testing that we are aware of at this time:

-
-
-

Objectives

Subproject Objectives

The objectives of the performance testing subproject are to:

-
-
-

The reason for establishing a performance test project is to...

The performance test subproject intends to address these specific concerns:

-
-
-

Target Performance Metrics

<Company Short Name> has the following target performance metrics for the performance measurements in this project:

Performance Measurement	Type of Measurement	Target Metric	Current Metric	Description

Critical Success Factors

The critical success factors in meeting the goals stated in the subproject objectives section are:

- Strong executive sponsorship and management support of the project mission and project team
- Adequate project staffing for the expected goals and timeline to be met
- Clear roles and responsibilities defined for the project in order to assure accountability, ownership, and quality

- A committed and well-informed performance testing team leader and project team that have a thorough understanding of the project goals and milestones as well as the relationship to the main implementation project
- A comprehensive project workplan and Project Management Plan
- A defined and maintained project infrastructure throughout the project
- A thorough understanding of known project risks and assumptions by business and project management

Approach

Performance Testing Process

The method chosen for this subproject adheres to the *Performance Testing* process within Oracle's Application Implementation Method.

Oracle Method provides a full life-cycle, service-integration methodology for Oracle Services' lines of business. For each line of business, Oracle Method provides a customizable project Workplan (including estimating factors and required project roles) with supporting guidelines. This methodology standardizes, integrates, and optimizes lines of business from strategic planning to production support. Oracle Method provides a consistent, deliverable-based approach to all of Oracle Services' lines of business.

Oracle Method Components

The lines of business listed below are the parts of Oracle Method that we will utilize for this project are:

- Application Implementation Method (AIM)
- Project Management Method (PJM)

The specific tasks defined in Oracle Method documentation for this project exist in the Project Workplan, together with any project specific additions. Work Management using the Workplan exists in the Project Management Plan.

AIM Phases

Oracle's Application Implementation Method (AIM) is being used on this project. The method covers the following management and control phases of the project life cycle:

- Definition *
- Operations Analysis *
- Solution Design *
- Build *
- Transition
- Production

Only the phases marked with an asterisk (*) have tasks relevant to the AIM Performance Testing process.

Key Deliverables

This section lists the tasks and deliverables to be produced during this architecture project and maps these tasks and deliverables to the AIM phase. The tasks and deliverables of this process are as follows:

ID	AIM Phase	Task Name	Deliverable	Review Type	Sign-Off
PT.010	Definition	Define Performance Testing Strategy	Performance Testing Strategy	W	PS
PT.020	Operations Analysis	Identify Performance Test Scenarios	Performance Test Scenarios	I	PM
PT.030	Operations Analysis	Identify Performance Test Transaction Models	Performance Test Transaction Models	I	PM
PT.040	Solution Design	Create Performance Test Scripts	Performance Test Scripts	I	TL
PT.050	Solution Design	Design Performance Test Transaction Programs	Performance Test Transaction Program Designs	W	TL
PT.060	Solution Design	Design Performance Test Data	Performance Test Data Design	I	TL
PT.070	Solution Design	Design Test Database Load Programs	Performance Test Database Load Program Designs	W	TL
PT.080	Build	Create Test Transaction Programs	Performance Test Transaction Programs	W	TL
PT.090	Build	Create Test Database Load Programs	Performance Test Database Load Programs	W	TL
PT.100	Build	Construct Performance Test Database	Performance Test Database	I	TL
PT.110	Build	Prepare Performance Test Environment	Performance Test Environment	I	TL
PT.120	Build	Execute Performance Test	Performance Test Results	Working deliverable-review not required	N/A
PT.130	Build	Create Performance Test Report	Performance Test Report	W	PS

Key: Review Type

I = Inspection, W = Walk through, P = Peer Review/Desk check

Sign-off

TL = Team Leader, PS = Project Sponsor, PM = Project Manager

Benefits

Listed below are potential benefits <Company Short Name> may recognize from this project by following the proven Performance Testing process included in AIM:

- structured top-down methodology for isolating performance issues
- approach links the identification of performance issues and bottlenecks to definition the development of a performance test
- relate performance test to the real production system environment for correct interpretation of results

- flexible process to accommodate automated or manual performance tests and a wide spectrum of scope from the entire system down to individual modules
- empirical technique for performance quality management in projects, deriving results that can be used for system tuning, capacity planning, and optimal setup of applications
- reusable test suite that can be utilized for future performance regression testing prior to introducing new software or hardware reconfiguration into the production system

Glossary

Benchmark Test

A test that measures the limiting performance metrics for a selected configuration and prescribed set of application test functions, as one or more *test parameters* will vary towards some limiting value. Is usually concerned with testing a small number of special transactions in a highly tuned environment, rather than simulating the complex business processing of an enterprise in live production. In the case of a published benchmark, an impartial external agency (such as TPC benchmarks) prescribes the test transactions.

Performance Test

A test which measures performance metrics for a hardware/software configuration that is processing a set of application test functions subject to a single set of *test parameters*. Usually the design of the test is to mimic a particular production processing environment. It differs from a *benchmark* in that a *benchmark* is less likely to consider simulating a production environment.

Performance Test Scenario

A point-in-time sample of the processing environment to be tested. Each scenario will in general include users working in multiple applications, with on-line and/or batch processing being performed.

Test Cycle

An instance of a test *transaction model* with a specific set of *test parameters* or a specific hardware configuration. For example, by varying the number of simulated users in multiple cycles, it is possible to determine the capacity of the configuration under test.

Test Parameters

Test parameters are the set of test variables that may alter during a series of *test cycles*, possibly for *benchmarking* purposes. Test parameters can change without altering the hardware/software configuration or the mix of transactions that comprise the overall application test functions. Examples are the total number of users being simulated in the test or the number of CPU slots utilized in a server.

Test Scenario

A description of the processing situations (point-in-time processing snapshots) in the future production system where there will be critical performance requirements. These performance *test scenarios* form the basis of models of the system transactions that make up the test.

Test Script

A program created to execute all or part of a specific *Test User Role*. If the *test user role* models a real application process flow in a *test scenario*, a component test script will cause the execution of transactions for a realistic process flow.

Test User Role

A set of transactions associated with a defined group of application users in the *Performance Test*. A *test user role* may include a set of transactions that attempt to model a 'real' application user process active during a *test scenario*. In some cases, a test user role facilitates segregation of the total active process flows, using a heuristic transaction grouping scheme. Transaction(s) process for each *test user role* through the execution of one or more unique *test scripts*.

Transaction Model

A specific mix of transactions for a particular *test scenario*. Each *test scenario* is supported by one or more *transaction models*.

Subproject Policies and Procedures

This section describes the policies and procedures we expect to follow for the Performance Testing subproject component of the <Project Name> project.

Relationship to Main Project Policies and Procedures

The performance testing subproject will use the same standards and procedures as the main project in the following areas:

- Project Management Plan
- Issue Management and Resolution
- Change Management
- Reporting Format
- Reporting Relationship to Main Project
- Acceptance
- Project Policies and Procedures
- Subproject Team Meetings
- Logistics and Administrative Support

Project Management Plan

The <Project Name> Project Management Plan includes the processes for Project Management and Quality Management of the main implementation project. The Project Management Plan includes the following sections:

- Introduction
- Scope
- Objectives
- Approach
- Project Tasks, Deliverables, and Milestones
- Control and Reporting
- Work Management
- Resource Management
- Quality Management
- Configuration Management

The Project Management Plan defines the detailed standards and procedures for use on the main project. The performance testing subproject will adhere to these standards also.

Issue Management Procedures

We will track and manage all performance testing issues as part of the main project level implementation process.

Issue Resolution

All performance testing issue resolution will utilize the main project issue resolution process.

Change Management

All performance testing change management requests will escalate up to the main project manager and gain resolution according to project wide change management standards.

Reporting Format

The format of formal and informal reports from members of the subproject team will be identical to that defined for the main project.

Reporting Relationship to Main Project

The organization chart, showing the reporting responsibilities of the members of the subproject team, will be prepared by <Performance Testing Team Leader>.

Acceptance

The Performance Testing subproject will go through acceptance by <Company Long Name> after completion of all deliverables. The procedure to be used for this is as follows:

-
-
-

On completion of Acceptance, we ask that the project sponsor sign an Acceptance Certificate as a record of the successful completion of the project and its defined scope.

Subproject Team Meetings

The Performance Testing team leader will call formal subproject meetings on a regular basis, prepare agendas, record any decisions, and delegate actions as defined in the Project Management Plan.

These meetings will occur weekly or biweekly...

Meeting Attendees (Subproject Role, Organization):

- 1.
- 2.
- 3.

Logistics and Administrative Support

The logistical support needed for this subproject will be similar to that required for other parts of the main project. Issues and policies regarding the logistical support will escalate up to the main project manager and resolution will occur according to project wide standards. The subproject will have availability to the same administrative support level as any other project group within the overall project.

Dependencies to Main Project and Other Subprojects

The performance testing process interacts with the following other teams in the main project and subprojects

- <Company Short Name> business requirements definition and mapping teams
- <Company Short Name> data conversion teams
- <Company Short Name> application and technical architecture team
- <Company Short Name> module design and build teams
- Project managers responsible for staffing the performance testing project

The dependencies of the Performance Testing subproject on other Application Implementation Method processes are:

- Application and technical architecture for the technical configuration of the new architecture and to feed back information about the performance of the architecture
- Business requirements definition and mapping for the new system processing functions and application setup data and to feed back information about the performance of the application configurations

- Data conversion database or data conversion programs
- Module Design and Build for database extensions and custom modules that need to be included in the performance testing, and to feed back performance measurements into the designs

Technical Background

The technical circumstances of the <Project Name> project will affect the scope of the performance testing process. Detailed below are the technical essentials of the project.

Implementation Sites

The project is implementing Oracle Applications in ... sites. These sites...

Technical Architecture

The technical architecture of the project is/is likely to be...

The new Oracle Applications systems will process on the following platforms/environments:

- Hardware versions:
- Operating system versions:
- Software versions:

The legacy applications in <Company Short Name>'s information systems architecture are running on the following platform/environment:

-
-
-

Data Conversion

The data converted in the project will be...

Environment Requirements for Subproject

This section summarizes the environments required to support the Performance Test process in the <Project Name>. It describes systems needed by the project team to carry out project activities.

The table below summarizes the application environments required to support the project team during the testing. If the performance test is happening after the implementation has gone live (for example to assess architecture or process changes), then it may be necessary to use the production environment for non-intrusive information gathering. The table below shows the correlation between expected environment usage and milestone task.

Milestone Task	Demo Database (DEMO)	Master Setup Data SETUP)	Development (DEV)	Performance Test (PERF)
Identify Performance Test Scenarios	✓		✓	
Identify Performance Test Transaction Models	✓		✓	
Create Performance Test Scripts	✓	✓	✓	
Design Performance Test Data	✓	✓	✓	
Create Performance Test Database		✓	✓	
Prepare Performance Test Environment				✓
Develop Performance Test Transaction Programs			✓	
Execute Performance Test				✓
Create Performance Test Report				✓

Technical Strategy

Description

Transaction Models

Transaction Execution

Database Construction

Performance Monitoring

During the test, we gather the following performance metrics:

Metric	Tool	Method
Response Times in transaction flows		
CPU Utilization		
Database Server Memory		
Disk I/O		
Oracle Data Dictionary		
Tier-to-Tier network traffic		

Test Execution

The performance testing subproject will employ the following test execution methods:

- 1.
- 2.
- 3.

Test Refresh Procedures

The Performance Test for <Company Short Name> executes in cycles, where a cycle is a particular test simulation run with pre-defined test parameters or conditions. Test refresh procedures execute in order to reset the test environment back to this initial baseline condition. These serve to reset the test environment back after the test cycle simulation runs increase the data volumes in the Performance Test database.

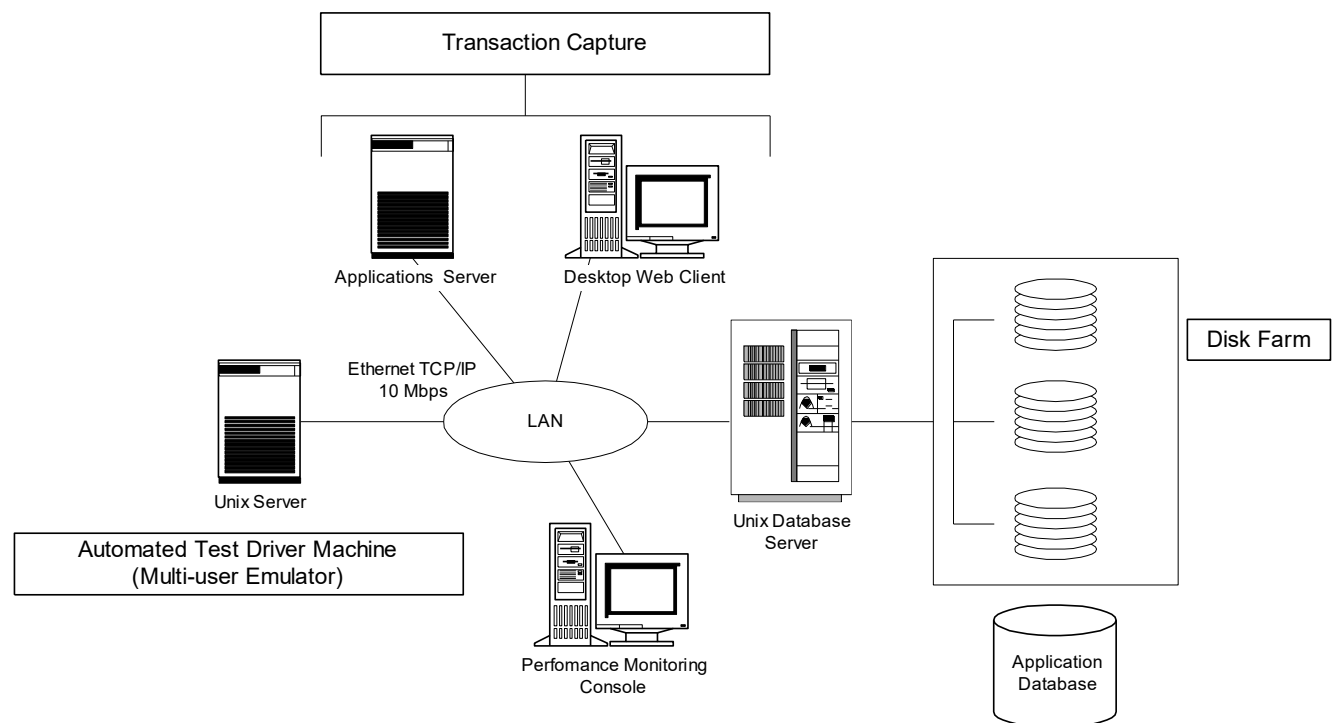
The refresh procedure is ...

Technical Architecture

The technical architectures documented here are the product of definition and design work elsewhere. At most, the Performance Test will help to select an optimal performing architecture from several choices, but it does not constitute a means of selecting and defining the architecture options in the first place.

Description

Diagram



Resource Requirements

This section summarizes the hardware, software, and staff resources required to support the Performance Test of Oracle Applications at <Company Long Name>. It describes systems needed by the project team to carry out project activities. This document serves as a planning document during the initial stages of the project to insure that the required environment is in place as each stage of project commences.

Software

The application software used as part of this project includes:

-
-
-

Hardware Environment

The hardware and operating system software used as part of this project include:

-
-
-

Networks

The network connection required for this project is:

-
-
-

Hardware and Software Delivery Schedule

New hardware and software to support the Performance Test is subject to the following delivery and installation schedule:

Vendor	Model/ Package	Description	Purpose	Scheduled Delivery	Delivered

Staff Resources

The staff involved with this project must have background, experience, and training in the following areas:

-
-
-

Below is a list of skills which the current performance testing project team does not fulfill:

-
-
-

Tool Requirements

The table below summarizes the tool requirements of significant tasks throughout the project:

Project Task	Applications	Word Processing	Project Library	Diagramming	Automated Testing Tool	Code Source Control	Performance Monitoring
Prepare Performance Testing Strategy		✓	✓	✓			
Identify Performance Test Scenarios		✓	✓				
Identify Performance Test Transaction Models	✓	✓	✓				
Create Performance Test Scripts	✓	✓	✓	✓			
Design Test Transaction Programs	✓	✓	✓		✓		
Design Performance Test Data	✓	✓	✓				
Design Test Database Load Programs	✓	✓	✓				
Create Performance Test Transaction Programs	✓	✓			✓	✓	
Create Test Database Load Programs	✓	✓	✓			✓	
Construct Performance Test Database	✓	✓	✓			✓	
Prepare Performance Test Environment	✓	✓	✓		✓	✓	✓
Execute Performance Test	✓	✓	✓		✓	✓	✓
Create Performance Test Report		✓	✓	✓			

The software selected/required for each of the above is as follows:

Applications: Oracle Applications

Source Control:

Word Processing: Microsoft Word

Diagramming: VISIO
Microsoft PowerPoint
(Tool dependent on type of diagram)

Automated Testing Tool:

Application Development

Automated Performance Testing

Performance Monitoring

Other System Management Tools

Risks

The performance testing strategy described here carries the following risks:

- 1.
- 2.
- 3.

Each item has an individual detailed description below:

Open and Closed Issues for this Deliverable

Open Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date

Closed Issues

ID	Issue	Resolution	Responsibility	Target Date	Impact Date